

Pricing · Curve Construction · Carry & Roll-Down Strategies

A comprehensive reference for rates & treasury practitioners

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1. Interest Rate Swaps (IRS)

An interest rate swap is a bilateral OTC contract in which two counterparties agree to exchange periodic cash flows calculated on a notional principal amount. One leg pays a fixed rate; the other pays a floating rate resetting periodically to a benchmark such as SOFR, €STR, SONIA, TONAR or CORRA. The notional itself is never exchanged. IRS are the most liquid derivative instruments in the world, with outstanding notional exceeding USD 400 trillion, and serve as the primary tool for managing duration, funding costs, and interest rate risk across all sectors of the capital markets.

1.1 Definition and Market Conventions

A vanilla fixed-for-floating swap is characterised by: the notional N , the tenor T , the fixed rate K (also called the swap rate or par rate), the floating index and its reset/payment frequency, the day count conventions on each leg, and the business day adjustment rules. Key conventions by currency:

Currency	Float index	Float freq.	Fixed freq.	Fixed DCC	Float DCC
USD	SOFR (compounded)	Annual	Semi-annual	30/360	Act/360
EUR	€STR (compounded)	Annual	Annual	30/360	Act/360
GBP	SONIA (compounded)	Annual	Semi-annual	Act/365	Act/365
JPY	TONAR (compounded)	Semi-annual	Semi-annual	Act/365	Act/365
CAD	CORRA (compounded)	Semi-annual	Semi-annual	Act/365	Act/365

Note: Post-LIBOR transition: all major G10 currencies have migrated to overnight RFRs (Risk-Free Rates) compounded in arrears. Term SOFR and Term €STR exist but are restricted in use by regulators.

1.2 Cash Flow Mechanics

Consider a T -year receiver IRS (receive fixed K , pay floating) on notional N . On each payment date t_i :

- Fixed leg receipt: $C^{\text{fix}}(i) = N \times K \times \text{dcf}^{\text{fix}}(t_{i-1}, t_i)$
- Float leg payment: $C^{\text{flt}}(i) = N \times R_i \times \text{dcf}^{\text{flt}}(t_{i-1}, t_i)$

where R_i is the compounded RFR rate observed over the accrual period $[t_{i-1}, t_i]$, and dcf denotes the relevant day count fraction. The net cash flow at each payment date is the algebraic sum of the two legs.

1.3 Pricing and Present Value

At inception, a par swap has zero NPV — the fixed rate K is set such that the present value of the fixed leg equals the present value of the floating leg. Using the OIS discount curve $P(0, t)$ (the discount factor from today to time t):

$$PV^{\text{fix}} = N \times K \times \sum \text{dcf}^{\text{fix}}(t_{i-1}, t_i) \times P(0, t_i)$$

$$PV^{\text{flt}} \approx N \times [P(0, t_0) - P(0, t_n)] \quad (\text{OIS float, no spread})$$

Setting $PV^{\text{fix}} = PV^{\text{flt}}$ and solving for K yields the **par swap rate**:

$$K = SR(0, T) = [P(0, 0) - P(0, T)] / \sum \text{dcf}_i \times P(0, t_i) = [1 - P(0, T)] / A(0, T)$$

where $A(0,T) = \sum dcf_i \times P(0, t_i)$ is the **annuity factor** (also called the level or DV01 basis). For an off-market swap struck at K with current par rate $SR(0,T)$, the NPV for the receiver is:

$$NPV^{\text{receiver}} = N \times (K - SR(0,T)) \times A(0,T)$$

Mark-to-Market intuition

If rates rise after you enter a receiver swap, $SR(0,T) > K$ and your swap has negative MTM — you are receiving below-market fixed. The loss is proportional to the rate move and the annuity. This is the core of duration/DV01 risk.

1.4 DV01, Duration and Risk Measures

The **DV01** (Dollar Value of 01) measures the change in NPV for a 1 basis point parallel shift in the swap curve. For a receiver swap:

$$DV01 = -\partial NPV / \partial r \approx N \times A(0,T) \times 0.0001$$

For a T -year swap on 10m notional, $DV01 \approx T \times 92$ USD (a useful approximation). More precisely, the **modified duration** of the fixed leg equals its Macaulay duration divided by $(1 + r/f)$ where f is the payment frequency. The swap's overall duration $\approx$ duration of fixed leg – duration of floating leg. Since the float leg resets at par each period, its duration $\approx$ time to next reset (typically 0.25–0.5 years), so the net duration $\approx$ duration of the fixed leg.

Tenor	Approx DV01 (10m notional)	Approx Duration (yrs)
2Y	~1,840	~1.9
5Y	~4,600	~4.7
10Y	~9,200	~8.9
20Y	~17,200	~15.8
30Y	~23,000	~20.1

1.5 Forward Rates and the Floating Leg

The floating leg can be valued by replacing each unknown fixing R_i with its forward rate $F(t_{i-1}, t_i)$ implied by the projection curve. The forward rate between dates T_a and T_b is:

$$F(T_a, T_b) = [P(0, T_a) / P(0, T_b) - 1] / dcf(T_a, T_b)$$

For a **forward-starting swap** $SR(T_a, T_b)$, the par rate is simply the rate that makes the forward annuity equal to the difference in discount factors:

$$SR(T_a, T_b) = [P(0, T_a) - P(0, T_b)] / A(T_a, T_b)$$

This is fundamental for carry/roll analysis: the **6M5Y forward swap rate** $SR(6M, 5Y)$ tells us at what rate the market is pricing a 5Y swap starting in 6 months, and is directly observable from the curve today.

2. Cross-Currency Swaps (XCCY)

A cross-currency swap (XCCY swap or CCS) is an agreement between two parties to exchange interest payments — and typically notional amounts — denominated in two different currencies over a specified period. They are the primary instrument used to transform funding from one currency to another, hedging both interest rate risk and FX risk simultaneously. Banks, sovereigns, supranationals and corporates use them extensively for synthetic foreign-currency issuance, balance sheet optimisation and regulatory liquidity management (LCR, NSFR).

2.1 Structure and Typology

The three main structures encountered in practice:

- **Fixed-Fixed XCCY:** both legs pay fixed rates in their respective currencies. Common in long-dated bond issuance hedges.
- **Fixed-Float (Coupon swap):** one leg is fixed, the other floating. Used when the issuer wants to convert fixed-rate debt into floating in the other currency.
- **Float-Float (Basis swap):** both legs pay floating rates (OIS + basis spread on one leg). This is the most liquid form for tenors beyond 1 year and is the instrument through which the xccy basis is directly observable.

In all structures, **notional principal is exchanged at inception and maturity** at the prevailing spot FX rate — a critical difference from single-currency IRS. In **Mark-to-Market (MtM) XCCY swaps**, the notional on the non-USD leg is reset periodically (typically quarterly or annually) to reflect the current FX rate, with a compensating cash flow paid. This eliminates FX settlement risk on the notional and is the standard in the interdealer market.

2.2 The XCCY Basis and CIP Deviations

In a textbook world with no frictions, **Covered Interest Parity (CIP)** holds:

$$F(0,T) / S_0 = (1 + r_{\text{USD}})^T / (1 + r_{\text{EUR}})^T$$

where S_0 is the spot rate, $F(0,T)$ the forward, and r the risk-free rates. Under CIP, a EUR investor borrowing EUR, swapping to USD via FX swap, and investing in USD should earn exactly the EUR risk-free rate — no arbitrage. Post-2008, however, CIP *systematically fails*. The **xccy basis** is the spread added to one leg (typically the non-USD leg) to restore equilibrium:

$$\text{Basis}(T) = r_{\text{OIS}}^{\text{USD}} - r_{\text{OIS}}^{\text{EUR}} - (F - S)/S/T \quad [\text{annualised, in bps}]$$

Structural drivers of persistent negative basis (non-USD pays basis to USD):

- Excess demand for USD funding from non-US banks (regulatory LCR, dollar asset growth)
- Balance sheet cost of providing USD via FX swap (SA-CCR, leverage ratio)
- Segmented investor base — EUR/JPY investors cannot freely arbitrage
- Counterparty and credit risk in the FX swap market

Pair	Typical 5Y basis (bps)	Key driver
EUR/USD	–20 to –30	European bank USD demand, LCR
JPY/USD	–50 to –80	JGB investor USD hedging, BoJ policy
GBP/USD	–15 to –25	UK bank funding, Gilt hedging

CAD/USD	–8 to –15	Relatively low — proximity to US market
AUD/USD	–30 to –50	High demand for AUD fixed income from US investors

2.3 Pricing a Mark-to-Market XCCY Swap

Consider a MtM XCCY swap: receive EUR fixed K_{EUR} , pay USD floating (SOFR flat), exchange notionals at start ($N_{USD} = S_0 \times N_{EUR}$) and maturity. The PV of each leg (from the EUR receiver's perspective):

$$PV^{EUR\text{ fix}} = N_{EUR} \times K_{EUR} \times \sum dcf_i \times P_{EUR}(0, t_i) + N_{EUR} \times P_{EUR}(0, T)$$

$$PV^{USD\text{ flt}} = N_{USD} / S_0 \times [1 + \text{spread} \times \sum dcf_i \times P_{USD}(0, t_i)]$$

At par (NPV=0), the **xcy par rate** incorporates both the interest rate differential and the basis. In a float-float basis swap, the spread (basis) is quoted directly on the non-USD leg. For a EUR receiver of float+basis vs USD SOFR flat:

$$\text{Basis}(T) = [P_{USD}(0, T) - P_{EUR}(0, T) \times S_0 / F(0, T)] / A_{EUR}(0, T) \text{ [simplified]}$$

Synthetic funding decomposition

A EUR issuer can synthetically issue USD debt by: (1) issuing EUR bonds at K_{EUR} , (2) entering a receive-EUR-fixed / pay-USD-floating XCCY swap. All-in USD cost = USD SOFR + xccy basis + basis vs IRS differential. This is directly comparable to USD FRN issuance cost.

2.4 FX Swap Decomposition

A short-dated XCCY swap (< 1 year) is economically equivalent to an **FX swap** — a simultaneous spot purchase and forward sale of currency. The FX swap implied USD rate is:

$$r_{\text{implied}}^{USD} = r^{EUR} + (F - S) / S / T = r^{EUR} + f_{EUR/USD}$$

where $f_{EUR/USD}$ is the FX forward premium/discount annualised. The FX swap market (rolling up to 1Y) is the deepest and most liquid segment. Beyond 1Y, MtM XCCY basis swaps take over. The basis measured in both markets should be consistent at the 1Y point — any gap represents arbitrage (subject to balance sheet cost).

2.5 Risk Measures for XCCY Swaps

The key Greeks for a XCCY swap position:

- **DV01 (rate sensitivity):** sensitivity to a 1bp parallel shift in the domestic rate curve. For receive-fixed EUR XCCY, this is similar to an EUR IRS DV01. The floating USD leg has negligible DV01 (resets to par at each period).
- **Cross-gamma / basis DV01:** sensitivity to a 1bp change in the xccy basis. For a T-year basis swap, basis DV01 $\approx N \times A_{EUR}(0, T) \times 0.0001$.
- **FX delta:** sensitivity to a 1% move in the spot FX rate. Arises from the notional exchange at maturity. In MtM structures this is largely neutralised by periodic notional resets, but residual FX delta remains intra-period.
- **CVA / FVA:** for long-dated XCCY swaps, counterparty exposure (PFE, EAD) and funding value adjustment are material, especially given notional exchanges and large MTM swings driven by FX moves.

3. Swap Curve Construction

Modern swap curve construction operates within a **multi-curve framework**: separate curves are built for discounting (OIS) and for projecting forward rates (tenor-specific projection curves). This replaced the single-curve approach after 2008 when LIBOR-OIS spreads widened dramatically, revealing that LIBOR-based forward rates carried credit risk not reflected in risk-free discounting.

3.1 Instruments Used for Bootstrapping

The curve is built from a set of liquid market instruments at increasing tenors. Each instrument contributes one node to the discount curve:

Segment	Instruments	Typical tenors	Observable quote
Short end (< 1M)	Central bank rate, Overnight deposit	O/N, T/N, S/N	Rate (%)
Money market (1M–1Y)	OIS swaps, FX swaps	1W, 1M, 3M, 6M, 1Y	Rate (%)
Belly (1Y–5Y)	IRS (par swaps)	2Y, 3Y, 4Y, 5Y	Par rate (%)
Long end (5Y–30Y+)	IRS (par swaps)	7Y, 10Y, 15Y, 20Y, 30Y	Par rate (%)

3.2 OIS Discounting and the Multi-Curve Framework

Under the multi-curve framework, two curves are constructed simultaneously:

- **OIS discount curve $P_{OIS}(0,t)$** : built from overnight index swaps (SOFR swaps, €STR swaps, SONIA swaps). Used to discount all cash flows regardless of the floating index on the swap. This reflects collateralised (CSA) funding costs.
- **Projection curves $P_{idx}(0,t)$** : built from instruments referencing the specific floating index (e.g., 3M SOFR, 6M EURIBOR — legacy). Used to project forward floating cash flows. Post-transition these collapse towards the OIS curve with only a small tenor basis spread.

The bootstrapping procedure solves for discount factors iteratively. At each tenor node T_n , the par rate $SR(0, T_n)$ is known from the market. Given all discount factors $P(0, T_1) \dots P(0, T_{n-1})$ already solved, the new discount factor is:

$$P(0, T_n) = [1 - SR_n \times \sum_{i=1}^{n-1} \text{dcf}_i \times P(0, T_i)] / [1 + SR_n \times \text{dcf}_n]$$

3.3 Interpolation Methods

Between known tenor nodes, discount factors (or zero rates / instantaneous forwards) must be interpolated. The choice of method matters significantly for forward rate stability and hedging accuracy:

Method	Applied to	Properties	Used for
Linear	Zero rates	Simple, produces jagged forwards	Quick approximation
Log-linear	Discount factors	Positive forwards guaranteed	Market standard (short end)
Cubic spline	Zero rates or forwards	Smooth, but can oscillate	Legacy systems

Natural cubic spline (monotone)	Instantaneous forwards	Smooth and positive forwards	Industry best practice
Hermite / tension spline	Instantaneous forwards	User-controlled tension, very smooth	Dealer systems

Note: The standard in most dealer systems is log-linear on discount factors for the short end (< 1Y) and monotone cubic Hermite interpolation on instantaneous forwards for the long end. Discontinuous forwards at node points are a known artefact of piecewise methods.

3.4 Tenor Basis and Spread Curves

In the post-RFR world, tenor basis (e.g., 1M SOFR vs 3M SOFR) is small but non-zero and must be modelled separately. The **tenor basis spread** $B(T_a, T_b)$ for tenor a vs tenor b satisfies:

$$F_a(T_{i-1}, T_i) = F_b(T_{i-1}, T_i) + B_{ab}(T_i)$$

In practice, a **basis swap curve** is built as a spread overlay on the OIS curve. The spread nodes are solved such that basis swaps (pay 3M index vs receive 6M index + spread) price at par. The 1Y tenor basis in EUR (€STR 1M vs 3M) currently trades around 1–3bps; historically LIBOR tenor bases were 20–30bps.

3.5 XCCY Basis Curve Construction

The XCCY basis curve for EUR/USD gives, at each tenor, the spread added to the EUR OIS floating leg such that the PV of EUR cash flows equals the PV of USD cash flows when exchanging at spot. It is bootstrapped from MtM XCCY swap quotes:

- **Step 1:** Take the USD OIS curve as given (built from SOFR swaps).
- **Step 2:** Take the EUR OIS curve as given (built from €STR swaps).
- **Step 3:** For each XCCY basis swap tenor T_n with quoted basis B_n , solve for the EUR discount factor $P_{\text{xcy}}(0, T_n)$ such that the swap prices at zero NPV when using B_n on the EUR leg.
- **Step 4:** The resulting **xcy discount curve** $P_{\text{xcy}}(0, t)$ is used to discount cross-currency cash flows when valuing a EUR-collateralised deal paying USD cash flows (the 'wrong-currency collateral' adjustment).

Practical note on collateral

The basis curve is tied to the collateral currency. A USD-collateralised EUR/USD XCCY swap uses the USD OIS curve to discount USD flows and the xccy curve to discount EUR flows. EUR-collateralised deals invert the logic. This is the 'cheapest-to-deliver collateral' optionality embedded in multi-currency CSAs.

4. Carry and Roll-Down Strategies

Carry and roll-down are the two components of the *time value* of a fixed income or derivative position — the return that accrues simply by holding the position, absent any market move. Together they quantify the **cushion** a position has against adverse rate or basis movements, and are central to the investment case for almost every rates trade.

4.1 Definitions: Carry vs Roll-Down

Carry — the certain component

Carry is the **known, contractually certain** income earned over a horizon H . For a spot-starting receiver IRS, carry for the first period is determined at inception because the floating fixing is already set. It requires **no assumption** about future rates. For forward-starting swaps, there is no carry — no payment occurs during the forward period.

Roll-Down — the conditional component

Roll-down is the **mark-to-market gain** that accrues if and only if the yield curve remains *exactly unchanged* as time passes. As a swap's remaining tenor shortens, the position is repriced at a different point on the curve. In an upward-sloping curve, shorter maturities have lower rates, so a receiver swap gains value as its tenor rolls down the curve. Roll is **not an expectation** — it is a scenario P&L conditional on static rates.

4.2 Carry on Spot IRS

For a spot-starting receiver swap on notional N , receiving fixed $SR(0,T)$ and paying the overnight/6M fixing $F(0,6M)$, the **forward value of carry** over a 6-month horizon is:

$$\text{Carry}(6M, FV) = N \times [SR(0,T) - F(0,6M)] / 2$$

Annualised and expressed in rate terms (bps), this is — following the Nordea convention — normalised by the annuity factor (dv01-equivalent) to give a basis-point yield:

$$\text{Carry}(6M) = [SR(0,T) - F(0,6M)] / dv01(\text{Swp}(6M, T-6M)) \quad [\text{Eq. 1a}]$$

A useful approximation valid for horizons $\leq$ the floating period:

$$\text{Carry}(6M) \approx SR(6M, T-6M) - SR(0,T) \quad [\text{Eq. 1b — approximate}]$$

Note: Eq. 1b uses the current implied future swap rate as a proxy. It is *erroneous* for horizons longer than the floating payment period (e.g., 12M carry on a 6M-resetting swap) because the second floating payment $F(6M,12M)$ is not yet fixed and cannot be treated as carry.

- **Positive carry** arises when the fixed rate exceeds the overnight fixing — i.e., the curve is not fully inverted at the front end relative to the swap maturity.
- **Negative carry** occurs in an inverted curve environment where short rates exceed long swap rates — the receiver pays away more on the float than it receives fixed.

4.3 Roll-Down on Spot and Forward IRS

Roll is computed by comparing today's swap rate at the original tenor to today's rate at the *shortened* tenor — i.e., where the swap will sit in H months if the curve is unchanged.

Spot-starting swap:

$$\text{Roll}(6M) = \text{SR}(0, T) - \text{SR}(0, T-6M) \quad [\text{Eq. 2a}]$$

Forward-starting swap Swp(F, T):

$$\text{Roll}(6M) = \text{SR}(F, T) - \text{SR}(0, T) \quad [\text{Eq. 2b}]$$

For Eq. 2b, as 6 months elapse, the 6M5Y swap becomes a spot 5Y swap. The roll is the difference between the forward swap rate today and the spot swap rate today. In an upward-sloping curve this is positive (forward rates > spot rates). For absolute P&L, multiply by $\text{dv01}(\text{Swp}(0, T-6M))$:

$$\text{Absolute Roll (USD)} = \text{Roll}(6M, \text{bps}) \times \text{DV01}(\text{Swp}(0, T-6M))$$

Instrument	6M Roll formula	Sign in normal upward curve
5Y spot receiver	$\text{SR}(0, 5Y) - \text{SR}(0, 4.5Y)$	Positive
10Y spot receiver	$\text{SR}(0, 10Y) - \text{SR}(0, 9.5Y)$	Positive
6M5Y fwd receiver	$\text{SR}(6M, 5Y) - \text{SR}(0, 5Y)$	Positive (fwd > spot)
2s5s flattener (DV01-neutral)	$[\text{SR}(6M, 5Y) - \text{SR}(0, 5Y)] - [\text{SR}(6M, 2Y) - \text{SR}(0, 2Y)]$	Positive if 5s roll > 2s roll

4.4 Carry and Roll on XCCY Basis Swaps

For XCCY basis swaps, the carry and roll framework adapts to basis dynamics. In a **receiver of basis** position (receive EUR float + basis, pay USD SOFR flat):

- **Carry:** the current basis level $B(T)$ is the known income for the first period (assuming the basis fixing is set). $\text{Carry} \approx B(T)$ bps annualised.
- **Roll:** as time passes and the remaining tenor shortens from T to $T-H$, the basis is repriced at $B(T-H)$. If the basis curve is downward sloping in absolute terms (less negative at shorter tenors, which is typical), the receiver of basis gains:

$$\text{Roll}_{\text{basis}}(H) = B(T-H) - B(T) \quad [\text{positive if basis less negative at } T-H]$$

The total carry+roll for a XCCY basis receiver over horizon H is:

$$\text{C+R}_{\text{XCCY}}(H) = B(T) + [B(T-H) - B(T)] = B(T-H)$$

Intuitively: if you receive basis and the basis curve is humped or upward-sloping (more negative at longer tenors), holding the position and rolling it shorter delivers a gain. The total C+R equals the basis you will be marked at in H months under a static curve assumption.

IRS overlay on XCCY

When a XCCY swap embeds a fixed EUR leg (receive EUR fixed vs pay USD float), the position has both an IRS component and a XCCY basis component. The total carry+roll is additive: $C+R_{\text{total}} = C+R_{\text{IRS(EUR)}} + C+R_{\text{basis}}$. The IRS carry/roll is driven by the EUR curve slope; the basis carry/roll by the XCCY basis curve slope. These can offset or reinforce each other and should be analysed separately.

4.5 Combined Carry+Roll and Break-Even Analysis

Carry and roll are **additive** because they have different assumptions: carry is certain (no curve assumption), roll is conditional (static curve). When combining, the total C+R in bps yield terms represents the **break-even adverse move** before the position loses money over the horizon H:

$$\text{Break-even rate move} = - (\text{Carry} + \text{Roll}) \text{ [in bps]}$$

If Total C+R = +15 bps on a 5Y receiver over 6 months, the position breaks even if rates rise by 15 bps over those 6 months. This is a key risk management input — it answers 'how much can the market move against me before I lose money on carry alone?'

Carry+Roll for a DV01-neutral spread trade (e.g., 2s5s flattener, receive 5Y / pay 2Y in DV01-neutral ratio r):

$$C+R_{\text{spread}} = C+R_{5Y} - C+R_{2Y} \text{ [without scaling, same DV01]}$$

These are directly additive because both legs share the same initial DV01. Absolute P&L = $C+R_{\text{spread}} \text{ (bps)} \times \text{DV01}$.

4.6 Tactical Trade Examples

The following examples illustrate practical carry+roll calculations. All rates are illustrative — replace with live BBG data in the dashboard.

Example A — EUR 10Y receiver spot (6M horizon)

Rate	Value
SR(0,10Y)	3.20%
SR(0,9.5Y)	3.13%
€STR fixing (6M)	3.92%
dv01(Swp(6M,9.5Y))	~8.74 per bp (10m)

- Roll(6M) = 3.20% – 3.13% = **+7.0 bps**
- Carry(6M) $\approx (3.20\% - 3.92\%) / \text{dv01} \times 0.9995 \approx$ **-8.2 bps** (negative carry — inverted curve)
- Total C+R = -8.2 + 7.0 = **-1.2 bps** — marginal negative, trade is breakeven at flat rates

Example B — EUR/USD 5Y XCCY basis receiver (6M horizon)

Rate	Value
B(5Y) EUR/USD basis	-22 bps
B(4.5Y) EUR/USD basis	-20 bps
IRS overlay carry (EUR 5Y vs €STR)	+8 bps

- Basis carry = -22 bps (you receive basis = negative number is income for receiver)
- Basis roll = $B(4.5Y) - B(5Y) = -20 - (-22) = +2$ bps
- IRS carry+roll on EUR fixed leg: **+12 bps** (carry 8 + roll 4)
- Total C+R = 22 (basis income) + 2 (basis roll) + 12 (IRS) = **+36 bps**

Example C — 2s10s EUR flattener, 6M5Y vs 6M2Y (DV01-neutral)

Rate	Value
SR(6M,10Y)	3.28%
SR(0,10Y)	3.20%
SR(6M,2Y)	3.55%
SR(0,2Y)	3.48%
DV01 (6M10Y, per 10m)	~8.84
Ratio (DV01-neutral)	~4.4x on 2Y

- Roll 10Y fwd receiver = $SR(6M,10Y) - SR(0,10Y) = 3.28 - 3.20 = +8$ bps
- Roll 2Y fwd payer = $SR(6M,2Y) - SR(0,2Y) = 3.55 - 3.48 = +7$ bps (cost)
- Net C+R = $8 - 7 = +1$ bp — modest but positive, modest carry cushion for the flattener

5. Summary Reference Tables

5.1 IRS vs XCCY — Key Characteristics

Feature	IRS	XCCY Basis Swap
Currencies	Single	Two
Notional exchange	None	At start and maturity (MtM: periodic resets)
Floating index	OIS (SOFR, €STR...)	OIS + xccy basis on non-USD leg
Main risk	Duration / DV01	Basis risk + FX delta + IRS
Primary use	Duration mgmt, funding	Cross-ccy funding, synthetic issuance
Curve sensitivity	Single-currency swap curve	Two OIS curves + xccy basis curve
Carry source	Fixed rate vs overnight fixing	Basis level + IRS differential
Roll-down driver	Yield curve slope	Basis curve slope + IRS roll

5.2 Carry and Roll — Formula Summary

Metric	Formula	Assumption
Carry (spot IRS, exact)	$\text{Carry} = [\text{SR}(0,T) - F(0,6M)] / \text{dv01}(\text{Swp}(6M, T-6M))$	None — certain payment
Carry (approximate)	$\text{Carry} \approx \text{SR}(6M, T-6M) - \text{SR}(0,T)$	Valid only for $H \leq \text{float period}$
Roll (spot IRS)	$\text{Roll} = \text{SR}(0,T) - \text{SR}(0, T-H)$	Curve unchanged
Roll (fwd IRS)	$\text{Roll} = \text{SR}(F,T) - \text{SR}(0,T)$	Curve unchanged
Basis carry (XCCY)	$\text{Carry}_{\text{basis}} = B(T)$	None — fixing set
Basis roll (XCCY)	$\text{Roll}_{\text{basis}} = B(T-H) - B(T)$	Basis curve unchanged
Break-even move	$\text{BE} = -(\text{Carry} + \text{Roll})$	Static curve, H horizon

5.3 Sign Convention for Carry+Roll

Position	Carry sign	Roll sign (normal curve)	Typical C+R sign
Receiver IRS (spot, normal curve)	+	+	++
Payer IRS (spot, normal curve)	–	–	—
Receiver IRS (inverted curve)	–	+	depends
Fwd receiver IRS (normal curve)	0	+	+
Receiver XCCY basis	+	+ (if basis less negative short)	++
Payer XCCY basis	–	–	—
2s10s flattener (recv 10Y, pay 2Y) (fwd)	0	+ if 10s roll > 2s roll	small +

5.4 Key Risk Parameters — Quick Reference

Parameter	Definition	Approx magnitude
DV01	P&L for +1bp parallel shift in rate	$\sim T \times 92$ USD per 10m notional
Basis DV01	P&L for +1bp shift in xccy basis	$\sim A(0,T) \times 0.0001 \times N$
Carry (6M, bps)	Certain income over 6M horizon	$SR(0,T) - \text{fixing [bps/2]}$
Roll (6M, bps)	MTM if curve unchanged over 6M	Curve slope per 6M step
Break-even (bps)	Rate rise that zeroes the total C+R	$= -(\text{Carry} + \text{Roll})$
PFE (XCCY)	Max expected exposure at 95th pctl	Driven by FX vol $\times T \times N$
SA-CCR EAD (XCCY)	Regulatory exposure for RWA	Add-on depends on tenor/FX vol

Further reading: Nordea Markets — 'Carry & Roll' (2013); Andersen & Piterbarg — 'Interest Rate Modeling' (3 vols); Brigo & Mercurio — 'Interest Rate Models — Theory and Practice'; BIS Working Paper 683 — 'CIP deviations, the dollar and frictions in international capital markets'.

Note: All numerical examples are illustrative only. Market rates change continuously. For live data, connect the accompanying Dash dashboard to Bloomberg (see `carry_roll_dashboard_bbg.py`). This document does not constitute investment advice.